

# Arvindh Arun

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[in linkedin.com/in/arvindh75](https://www.linkedin.com/in/arvindh75)   [twitter.com/arvindh\\_\\_a](https://twitter.com/arvindh__a)

## Education

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| <b>Current</b><br><b>Aug 2024</b>  | <b>ELLIS and IMPRS-IS PhD (University of Stuttgart &amp; University of Edinburgh)</b><br>PhD in Artificial Intelligence<br>‣ Advised by <a href="#">Prof. Steffen Staab</a> and <a href="#">Prof. Antonio Vergari</a><br>‣ Working on Foundation Models for Knowledge Graphs, LLM Evaluations, Graph-based Retrieval Augmented Generation for LLMs                                                             | <b>Stuttgart, Germany</b> |
| <b>May 2024</b><br><b>Aug 2019</b> | <b>International Institute of Information Technology (IIIT H)</b><br>B.Tech. + MS by Research, Computer Science & Engineering<br>‣ Cumulative GPA - 9.07/10.00<br>‣ Dean's Academic List - 2020-21 M (Top 10%), 2021-22 M (Top 15%), 2022-23 S (Top 5%)<br>‣ Dean's Research List - 2022-23 (Top 5%)<br>‣ Represented the College Football Team, 5x medals in Track & Field, 3x medals in Inter-house football | <b>Hyderabad, India</b>   |

## Experience

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| <b>Aug 2024</b><br><b>Jan 2024</b> | <b>Google DeepMind   Information Networks (InfoNet) Team</b><br><i>Student Research Collaborator</i><br>‣ Worked on leveraging topological information from user interactions to gather insight into the quality of user-generated content<br>‣ Extended Vietoris Rips Complex and Persistent Images to post profiling on social networks<br>‣ Introduced a new benchmark to evaluate controversy detection models mimicking the real-world data distribution                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Remote</b>           |
| <b>Jul 2023</b><br><b>May 2023</b> | <b>JPMorgan Chase   Corporate Investment Banking</b><br><i>Applied AI/ML Intern</i><br>‣ Worked on mining interesting insights from transactional databases for Root Cause Analysis.<br>‣ Utilized streaming frameworks such as PySpark to optimize the workflow.<br>‣ Proposed a new framework utilizing unsupervised GNNs inspired by the CTR prediction domain.<br>‣ Received a Pre-placement Full-time offer for my contributions.                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Bangalore, India</b> |
| <b>Aug 2022</b><br><b>Jun 2022</b> | <b>The Chinese University of Hong Kong   Department of Systems Engineering</b><br><i>Research Intern   Advisor: <a href="#">Prof. Hong Cheng</a></i><br>‣ Worked on the domain of Clustering/Community detection in Dynamic social networks and studying their properties.<br>‣ Focused on minimizing the time complexity and the space complexity of the existing real-time solutions.<br>‣ Explored various state-of-the-art solutions and built a visualization tool to study their efficacy.                                                                                                                                                                                                                                                                                                                                                                                   | <b>Hong Kong</b>        |
| <b>May 2024</b><br><b>May 2021</b> | <b>Precog Research Group [🌐]</b><br><i>Undergraduate Researcher   Advisor: <a href="#">Prof. Ponnurangam Kumaraguru "PK"</a></i><br>‣ Analyzing social networks and other large networks primarily using their network information.<br>‣ <b>Georgia Institute of Technology, Atlanta</b><br>Collaboration with <a href="#">Prof. Polo Chau</a> on analyzing the Quantitative and Qualitative impact of fraudulent behavior on social networks and their identification.<br>‣ <b>Adobe Research, Bangalore</b><br>Collaboration with <a href="#">Dr. Ramasuri Narayanam</a> on exploring the landscape of group fairness in unsupervised contrastive learning based GNNs.<br>‣ <b>Indiana University, Bloomington</b><br>Collaboration with <a href="#">Prof. Jisun An</a> on analyzing the changes to the hate speech dynamics after the liberalization of social media platforms. | <b>Hyderabad, India</b> |
| <b>May 2021</b><br><b>Jan 2021</b> | <b>NemoCare</b><br><i>Cross Platform Development Intern</i><br>‣ Built the first prototype of the Android and iOS app for this Shark Tank funded, Google accelerated, Asia Hardware Battle 2019-winning startup led by a Forbes 30 under 30 recipient.<br>‣ Integrated the app with their pre-existing AWS backend ecosystem and their hardware by leveraging Lambda and EC2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Bangalore, India</b> |

## Publications

- [9] **The Illusion of Diminishing Returns: Measuring Long Horizon Execution in LLMs** [PDF]  
Akshit Sinha\*, Arvindh Arun\*, Shashwat Goel\*, Steffen Staab, Jonas Geiping (\* = Equal Contribution)  
*Under Review* [Arxiv]
- [8] **SEMMA: A Semantic Aware Knowledge Graph Foundation Model** [PDF]  
Arvindh Arun, Sumit Kumar, Mojtaba Nayyeri, Bo Xiong, Ponnurangam Kumaraguru, Antonio Vergari and Steffen Staab  
*The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)* [EMNLP '25 (Main)]
- [7] **A Cognac shot to forget bad memories: Corrective Unlearning in GNNs** [PDF]  
Varshita Kolipaka, Akshit Sinha, Debangana Mishra, Sumit Kumar, Arvindh Arun\*, Shashwat Goel\*, and Ponnurangam Kumaraguru (\* = Equally Advising)  
*42<sup>nd</sup> International Conference on Machine Learning (ICML)* [ICML '25]
- [6] **Best Paper** **Topo Goes Political: TDA-Based Controversy Detection in Imbalanced Reddit Political Data** [PDF]  
Arvindh Arun\* Karuna K Chandra\*, Akshit Sinha, Balakumar Velayutham, Jashn Arora, Manish Jain, and Ponnurangam Kumaraguru (\* = Equal Contribution)  
*5<sup>th</sup> International Workshop on Computational Methods for Online Discourse Analysis @ WWW* [BeyondFacts @ WWW '25]
- [5] **Sanity Checks for Evaluating Graph Unlearning** [PDF]  
Varshita Kolipaka, Akshit Sinha, Debangana Mishra, Sumit Kumar, Arvindh Arun, Shashwat Goel, and Ponnurangam Kumaraguru  
*3<sup>rd</sup> Conference on Lifelong Learning Agents Workshop Track (CoLLAS)* [CoLLAS-W '24]
- [4] **X-posing Free Speech: Examining the Impact of Moderation Relaxation on Online Social Networks** [PDF]  
Arvindh Arun\*, Saurav Chhatani\*, Jisun An, and Ponnurangam Kumaraguru (\* = Equal Contribution)  
*8<sup>th</sup> Workshop on Online Abuse and Harms (WOAH) at NAACL* [WOAH @ NAACL '24]
- [3] **Oral** **CAFIN: Centrality Aware Fairness inducing IN-processing for Unsupervised Representation Learning on Graphs** [PDF | code]  
Arvindh Arun, Aakash Aanegola, Amul Agrawal, Ramasuri Narayanam, and Ponnurangam Kumaraguru  
*26<sup>th</sup> European Conference on Artificial Intelligence (ECAI)* [ECAI '23]
- [2] **Erasing Labor with Labor: Dark Patterns and Lockstep Behaviors on the Google Play Store** [PDF]  
Ashwin Singh, Arvindh Arun, Pulak Malhotra, Pooja Desur, Ayushi Jain, Duen Horng Chau, and Ponnurangam Kumaraguru  
*LBR Track at 33<sup>rd</sup> ACM Conference on Hypertext and Social Media (ACM HT)*  
**Media** Work covered by [Montreal AI Ethics Institute](#) [LBR @ HT '22]
- [1] **Ensembling Pre-Trained Language Models with Feature Engineering** [PDF | code]  
T.H. Arjun, Arvindh Arun, and Ponnurangam Kumaraguru  
*GermEval 2021 Shared Task on the Identification of Toxic, Engaging, and Fact-Claiming Comments* [GermEval '21]

## Honours, Awards and Test Scores

**Y Combinator AI Startup School, 2025** [🌐] Accepted to attend a funded 2-day AI startup school in the Bay Area.

**Google Research Week, 2024** [🌐] Accepted to attend a funded 3-day ML research event, hosted by Google Research India.

**Microsoft Research Travel Grant, 2023** [🌐] Awarded \$2500 for attending ECAI'23 held in Krakow, Poland.

**Stellantis NV Hackathon, 2022** Finished as a runner-up in both rounds, winning a cash prize worth 125,000 INR.

**Facebook Hackercup, 2022** Reached Round 2, finished Top 10% globally.

**GermEval, 2021** Ranked 6<sup>th</sup> globally on cumulative task rank.

**PicoCTF by CMU, 2021** Finished 102<sup>nd</sup>, Top 2% globally.

**HowzHack (DreamVu), 2019** Finished 2<sup>nd</sup> for the problem statement exploring the applications of computer vision for Football's Video Assistant Referees (VAR).

**UGEE, 2019** 20<sup>th</sup> rank in India.

**NextGenius Scholarship, 2019** Finalist (withdrew) selected for fully funded Undergraduate programme at University of The South, Tennessee, USA.

## Teaching

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- Foundations of Machine Learning (UoS)** *Teaching Assistant* April '25 - June '25
- > Introductory ML course for undergrads.
  - > Covered lectures on Linear Algebra.
- Deep Learning Lab (UoS)** *Course Co-instructor* Oct '24 - Feb '25
- > Introductory lab course for Deep Learning with a special focus on Knowledge Graphs, Diffusion models, and LLMs.
  - > Covered lectures on Graph Learning and Good practices in Deep Learning.
- Computational Social Science (IIITH - CS4.301)** *Teaching Assistant* Jan '23 - May '23
- > Course covers social computing concepts and social media analysis.
  - > Responsibilities included curating assignments, conducting weekly tutorials, and grading quizzes and assignment submissions.
- Data & Applications (IIITH - CS9.435)** *Teaching Assistant* Aug '22 - Dec '22
- > Course covers the basics of database systems and introduces SQL.
  - > Responsibilities included curating assignments, conducting weekly tutorials, and grading quizzes and assignment submissions.

## Academic Service

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**Reviewer** ACL ARR October '24, ICLR '25

## Selected Research Projects

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- Ensemble of Task-Specific Language Models for Brain Encoding** [[Arxiv](#) | [🔗](#)] Jan '23 - May '23  
*Cognitive Science and AI (CS9.432)*
- We improve the performance of brain encoders by creating an ensemble model out of 10 popular Language Models (2 syntactic and 8 semantic). Going by the hypothesis that different LMs capture different forms of semantics better, we beat the current baselines by 10% on average across all ROIs through our ensembling methods.
- Privacy Policy QnA bot** [[🔗](#)] Aug '22 - Dec '22  
*Online Privacy (CS4.407)*
- We fine-tuned the Jurassic-1 language model on a new dataset consisting of top Indian companies' Privacy Policies for QA and Summarization tasks. Also deployed a Telegram bot for easy access by connecting it with our hosted model.
- Wikipedia Open QnA** [[PDF](#) | [🔗](#)] Aug '21 - Dec '21  
*Advanced NLP (CS7.501)*
- We implemented the DrQA Retriever and Reader, and extracted features required to run the RankQA model. We also indexed a large Wikipedia data dump to train on and tested it on the popular SQuAD dataset.